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CAP2

CAPSTONE PROJECT 2

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Preface

The Capstone Project represents the culmination of a student's journey in Information Technology (IT) and related computing disciplines. It is an opportunity to transform knowledge into action, theory into practice, and ideas into tangible solutions that can make a real-world impact. Completing a capstone project challenges students to think critically, solve complex problems, and demonstrate the skills they have honed throughout their academic career.

This module, Capstone Project 2, is designed to guide students step by step through the process of planning, developing, documenting, and presenting their projects. It provides practical examples, templates, and best practices for each chapter of a capstone report, from the introduction and methodology to the results, discussions, and final recommendations. Students will also learn how to properly document their sources, apply academic writing standards, and use professional templates such as those from the Association for Computing Machinery (ACM).

Beyond technical guidance, this module encourages students to cultivate creativity, ethical research practices, and effective communication. It emphasizes the importance of requirement analysis, system design, software evaluation, and usability testing, ensuring that students produce projects that are not only functional but also innovative, user-friendly, and aligned with industry standards.

This module guide, students in gaining the confidence and skills to complete their capstone projects successfully, presenting work that is academically rigorous, professionally polished, and personally fulfilling. It is our hope that this module inspires students to embrace the challenges of their Capstone Project and to create IT solutions that can contribute meaningfully to society.

Introduction

The Capstone Project represents the pinnacle of a student's journey in Information Technology (IT) and Information Systems (IS), serving as a comprehensive platform to integrate theory, practice, and innovation. It challenges students to apply the knowledge and skills acquired throughout their academic program to design, develop, and evaluate functional IT solutions that address real-world problems.

This module, Capstone Project 2, provides students with a structured and practical guide to complete their projects successfully. It covers all essential aspects of project development, from conceptualization and requirement analysis to system design, software evaluation, and documentation. Additionally, it offers guidance on proper academic writing, source citation, and professional presentation, including the use of industry-standard templates such as the Association for Computing Machinery (ACM) format.

A key feature of this module is its alignment with CHED Memorandum Order (CMO) 25, series of 2015, which provides guidelines for the preparation of Capstone Projects and theses in Computing Studies. CMO 25 s.2015 emphasizes standardized approaches to research design, methodology, documentation, and evaluation, ensuring that student projects adhere to national quality standards. By following the CMO 25 framework, students are guided in conducting systematic research, applying rigorous software development practices, and producing results that are valid, reliable, and replicable.

Capstone projects are not merely academic exercises; they are opportunities for students to demonstrate critical thinking, creativity, and technical proficiency. This module emphasizes the importance of following established methodologies, adhering to software development and evaluation standards, and producing ethical, high-quality, and user-centered solutions. Through its lessons, students learn to conduct requirement analysis, apply research designs, implement testing strategies, and interpret results, all while maintaining academic integrity and professional rigor.

By utilizing this module, students are prepared to produce capstone projects that are academically sound, professionally polished, and compliant with CMO 25 s.2015. It is designed to inspire students to embrace challenges with diligence, innovation, and a commitment to excellence, bridging the gap between academic knowledge and industry-ready skills.

Lesson 1:

Introduction to Capstone Project Writing for Information Technology

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A. Learning Objectives

1. Define research and its significance in IT-related projects.
2. Explain the purpose of research in the context of solving real-world IT problems.
3. Identify the key characteristics that make research valid and reliable.
4. Differentiate between various types of research relevant to IT.
5. Describe common research methods and their application in IT projects.
6. Explain the concept of design in IT research and distinguish it from traditional research.
7. Differentiate between a research problem and a design problem in IT project development.

B. Content

Introduction

In Information Technology (IT) education, the Capstone Project represents the culmination of a student's learning journey, integrating skills and knowledge acquired over several years into a single, practical, and research-oriented work. Whether the output is a software application, a database system, or a web-based platform, it must be grounded in systematic research and design principles. This chapter introduces the foundations of research and design in the context of IT projects, equipping students with the understanding needed to plan and execute a successful capstone project.

The Meaning of Research

Research is the systematic process of collecting, analyzing, and interpreting information to increase understanding of a phenomenon or solve a specific problem. In IT, research goes beyond academic theory — it often focuses on developing new systems, improving existing technologies, or finding solutions to technical challenges.

Key points:
- Involves a structured approach and clear methodology.
- Requires critical thinking and evidence-based decision-making.
- Can be qualitative (descriptive, exploratory) or quantitative (statistical, measurable).

Example in IT: *Investigating the performance differences between relational and NoSQL databases for large-scale e-commerce applications.*

Purpose of Research

The purpose of research in IT capstone projects is to:

1. Generate New Knowledge – Provide innovative insights into technical processes, tools, or user behaviors.
2. Solve Problems – Address real-world IT challenges such as cybersecurity threats, system inefficiencies, or usability issues.
3. Guide Decision-Making – Support evidence-based development choices for system architecture, programming language selection, or deployment strategy.
4. Validate Hypotheses – Test whether a proposed system design or feature will perform as intended.

Characteristics of Research

Effective research should possess the following characteristics:

1. Systematic – Follows a step-by-step procedure.
2. Logical – Draws valid conclusions based on evidence.
3. Empirical – Based on direct or indirect observation and experience.

4. Replicable – Can be repeated to verify results.
5. Ethical – Follows guidelines for integrity, data privacy, and respect for participants.

Types of Research

Common research types applicable in IT projects include:

1. **Descriptive Research** – Describes characteristics of a system or user behavior.
   Example: *Analyzing user login frequency patterns in an online learning system*.
2. **Experimental Research** – Involves controlled testing to determine cause-effect relationships.
   Example: *Measuring the impact of a new encryption algorithm on system performance.*
3. **Applied Research** – Focuses on solving a specific, practical problem.
   Example: *Developing a chatbot to assist visually impaired users in navigating a website.*
4. **Developmental Research** – Involves designing, building, and testing a new IT product or system.
   Example: *Creating a mobile app for disaster response coordination.*

Research Methods

Research methods in IT typically fall into two main categories:

1. Quantitative Methods – Use measurable data to test hypotheses.
   - Surveys with numerical ratings
   - Performance benchmarking
   - Statistical analysis of system logs
2. Qualitative Methods – Focus on understanding experiences and perceptions.
   - Interviews with end-users
   - Observation of system use
   - Usability testing sessions
3. Mixed-Method Approach – Combines both to gain comprehensive insights.
   - **Convergent Parallel Design:** Quantitative and qualitative data are collected and analyzed simultaneously.
   - **Explanatory Sequential Design:** Quantitative data is collected and analyzed first, followed by qualitative data collection and analysis.
   - **Exploratory Sequential Design:** Qualitative data is collected and analyzed first, followed by quantitative data.

The Meaning of Design for IT Research

In IT, design refers to the planning and creation of a solution to a defined problem, focusing on functionality, usability, and technical performance.

1. **Research and Design Distinguished:**
   - Research: Seeks to generate knowledge or test theories.
     Example: Studying cloud computing adoption rates among small businesses.
   - Design: Seeks to produce a functional solution.
     Example: Creating a hybrid cloud architecture for a client.

   In a capstone project, both research and design often work together — research informs the design, and design outcomes can lead to further research.
2. **Research Problem vs Design Problem:**
   - Research Problem: A question about a knowledge gap that needs investigation.
     Example: 'What factors influence the load time of a mobile banking app?'
   - Design Problem: A practical challenge requiring a functional solution.
     Example: 'How can we redesign the mobile banking app to load in under 2 seconds?'

C. Summary

This chapter established the foundations of capstone project writing in IT, emphasizing the role of research and design. Research provides the evidence base for decision-making, while design translates findings into practical, innovative solutions. Understanding the differences between research and design, as well as between research and design problems, prepares students to create projects that are both academically rigorous and practically valuable.

D. Learning Assessment

Part I – Identification

1. The process of systematically collecting and analyzing information to increase understanding of a phenomenon is called:
   a) Hypothesis Testing
   b) Research
   c) Design Thinking
   d) Brainstorming

2. Which type of research focuses on solving a specific, real-world problem?
   a) Basic Research
   b) Applied Research
   c) Theoretical Research
   d) Historical Research

3. In IT, which research method typically uses interviews, observations, and case studies?
   a) Quantitative
   b) Qualitative

c) Experimental
d) Statistical

Part II – True or False

4. A research problem and a design problem are exactly the same.

5. Descriptive research in IT involves controlled testing of cause-and-effect relationships.

6. Research must be systematic, logical, empirical, replicable, and ethical.

Part III – Short Answer

7. Give one example of a research problem in IT.

8. Give one example of a design problem in IT.

9. State two main differences between research and design.

Part IV – Application

- Write a short paragraph (4–5 sentences) explaining how research methods can help improve the design of a mobile application.

E. References

- Creswell, J. W., & Creswell, J. D. (2018). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. SAGE Publications.
- Dawson, C. (2019). Projects in Computing and Information Systems: A Student's Guide. Pearson Education.
- Oates, B. J. (2006). Researching Information Systems and Computing. SAGE Publications.
- Robson, C., & McCartan, K. (2016). Real World Research. Wiley. Sommerville, I. (2016). Software Engineering. Pearson Education.

Lesson 2:

Writing the Component of Chapter 1

A. Learning Objectives

By the end of this module, students will be able to:

1. Differentiate between a thesis and a capstone project.
2. Identify the major fields under CMO 25, s. 2015 and their respective areas of study.
3. Familiarize themselves with the standard format of thesis and capstone projects.
4. Write the essential components of Chapter 1 such as the introduction, objectives, significance, scope and limitations, definition of terms, and theoretical/conceptual framework.
5. Apply academic writing skills in preparing the initial chapters of their research or project study.

B. Content

Introduction

Chapter 1 is the foundation of any thesis or capstone project. It introduces the study, defines its objectives, provides background information, and frames the research within theoretical and practical contexts. Writing Chapter 1 requires clarity, accuracy, and relevance, as it sets the tone and direction for the entire paper. This module provides a detailed guide to the components of Chapter 1, based on the Commission on Higher Education (CHED) Memorandum Order (CMO) No. 25, s. 2015, which outlines policies and standards for computing programs such as Computer Science, Information Technology, and Information Systems.

What is a Thesis?

A thesis is an academic research study undertaken by students to contribute new knowledge, validate existing theories, or propose solutions to problems through systematic research. In computing, a thesis often involves experimental studies, algorithm development, or evaluation of technological innovations.

Purpose: To generate new knowledge or validate a theory.
Output: A research paper with experiments, data analysis, and findings.
Example (Computing): A Machine Learning-Based Intrusion Detection System for Cloud Servers.

What is a Capstone Project?

A capstone project is a practical, project-based study where students integrate knowledge and skills to solve real-world problems. Unlike a thesis, it focuses on design, implementation, and evaluation rather than theoretical contributions.

Purpose: To showcase the application of knowledge and skills in solving industry-related problems.
Output: A system, software, or application with documentation.
Example (IT): Development of a Web-Based Employment Platform for Persons with Disabilities.

Topics Covered for Computing Studies Based on CMO 25, s.2015

CMO 25 outlines program outcomes and research areas for computing students.

- **COMPUTER SCIENCE**
    1. Current Computer Science Topics

- Software and Development Theory
- Mobile Computing Systems
- Software Extensions or Plug-ins
- Expert Systems and Decision Support System
- System Software (Software tools/utilities, interpreters, simulations, compilers, security aspects)
- Intelligent Systems Game
- Development Computer Vision
- Image/Signal Processing Natural
- Natural Language Processing
- Pattern recognition and data mining
- Bioinformatics Graphics
- Applications Cloud Computing
- Parallel computing
- Embedded systems
- Emerging Technologies

2. Foundation of Computer Science
   A. Automata and Formal Languages
   B. Data Structures and Algorithm Design and Analysis
   C. Web Semantics
   D. Coding theory
   E. Programming languages
   F. Visualization systems
   G. Computer and Architecture
   H. Modeling and Simulation
   I. Grid Computing
   J. Pervasive Computing

3. Human-Computer Interaction
   A. Usability
   B. Affective Computing
   C. Emphatic Computing

- **INFORMATION TECHNOLOGY**

1. Software Development
   - Software Customization
   - Information Systems Development for an Actual Client (with pilot testing)
   - Web Applications Development (with at least alpha testing on live servers)

    - Mobile Computing Systems
2. Multimedia Systems
    - Game Development
    - e-Learning Systems
    - Interactive Systems
    - Information Kiosks

- **NETWORK DESIGN AND IMPLEMENTATION AND SERVER FARM CONFIGURATION AND MANAGEMENT**
  1. IT Management
      - IT Strategic Plan for sufficiently complex enterprises
      - IT Security Analysis, Planning and Implementation
- **INFORMATION SYSTEMS**
  1. Software Development
      - Software Customization
      - Information Systems Development for an Actual Client
      - Web Applications Development
      - Mobile Computing Systems
  2. IS Planning
      - Enterprise Resource Plan
      - Information Systems Strategic Plan
  3. Analysis and Design of a sufficiently complex business system

Sample Thesis and Capstone Project Format (Based on CMO 25, s.2015)

CS Thesis (Foundation of Computer Science)	CS Thesis (Software Development)	IT / IS Capstone Project
• Title Page • Abstract • Table of Contents • List of Tables, Figures, Notations/Appendices	• Title Page • Abstract or Executive Summary • Table of Contents • List of Tables, Figures, Notations/Appendices	• Title Page • Executive Summary • Table of Contents • List of Tables, Figures, Notations/Appendices
Chapter I: Introduction		
• Introduction • Background of the Study • Statement of the Problem • Objectives of the Study	• Introduction • Project Context • Purpose and Description • Objectives of the Study • Scope and Limitations • Definition of Terms	• Introduction • Project Context • Purpose and • Description • Objectives of the Study • Scope and Limitations • Definition of Terms

• Significance of the Study • Scope and Limitations • Definition of Terms		
Chapter II: Review of Related Literature and Studies		
• Related Literature • Synthesis • Theoretical Background	• Related Literature • Synthesis • Technical Background	• Related Literature/System • Synthesis • Technical Background
Chapter III: Methodology		
• Proposed Solution to the Problem	• Design and Methodology	• Methodology • Requirements Analysis • Requirements Documentation • Design of the Software, System, Product and/or Process • Development and Testing • Description of the Prototype • Implementation Plan
Chapter IV: Results and Discussion		
• Results and Discussion • Summary of Findings	• Results and Discussion • Summary of Findings	• Results and Discussion • Summary of Findings
Chapter V: Conclusions and Recommendations		
• Conclusions • Recommendation	• Conclusions • Recommendation	• Conclusions • Recommendation
• Appendices	• Appendices	• Appendices

Writing a Good Introduction

The introduction provides an overview of the study. It should:

- Capture the interest of the reader.
- Present the context or background of the problem.
- Highlight the gap or need for the study.
- Clearly lead to the research objectives.

Tip: Start broad, then narrow down to the specific issue being addressed.

Formulating the Statement of the Objectives

Objectives explain what the study seeks to achieve.
They are classified as:
-**General Objective**: The overall aim of the study.
-**Specific Objectives**: Detailed goals that break down the general objective into measurable actions.

Example:
General Objective: To develop a mobile application for campus navigation.
Specific Objectives:
1. To design an interactive map for students.
2. To integrate real-time GPS navigation.
3. To evaluate usability through student feedback.

The Significance of the Study and the Purpose and Description

This section explains why the study is important and who will benefit from it.

Beneficiaries may include:
- Students
- Faculty and school administration
- Industry partners
- Future researchers

Example: This study will benefit students by providing a convenient tool for navigating campus facilities. It will also aid the administration in managing building and room information efficiently.

Scope and Limitation of the Study

- **Defines the boundaries of the study.**
  - Scope: What is covered (system features, population, timeframe).
  - Limitation: What is not covered due to constraints such as budget, resources, or time.

  Example:
  Scope: The system will include user login, appointment scheduling, and notifications.
  Limitation: The system will not include online payment processing.

Definition of Terms

Clarifies technical or specialized terms used in the study. Terms may be:
- Conceptual Definition: Based on literature.
- Operational Definition: Based on usage in the study.

Example:
Cloud Computing (Conceptual): The practice of using a network of remote servers hosted on the internet to store, manage, and process data.
Cloud Computing (Operational): In this study, it refers to the use of Amazon Web Services for hosting the proposed system.

The Theoretical and Conceptual Framework

- **Theoretical Framework:** Based on established theories that support the study. Example: Technology Acceptance Model (TAM), Agile Development Model.
- **Conceptual Framework:** The researcher's own model showing the flow of the study. Often illustrated with a diagram showing inputs, process, and outputs.

  Example (IPO Model):
  - Input: User requirements, hardware/software resources.
  - Process: System design, development, testing.
  - Output: Functional web-based employment system.

C. Summary

Chapter 1 provides the foundation of a thesis or capstone project. It introduces the study, defines objectives, describes its significance, establishes boundaries, and frames it in theory. Following the CHED CMO 25, s.2015 guidelines ensure standardization across computing programs.

D. Learning Assessment

1. Write a draft introduction for a capstone project titled: 'A Web-Based Appointment and Scheduling System for University Clinics.
2. Formulate a general objective and three specific objectives for the same project.
3. Identify at least three beneficiaries of the project and explain how they will benefit.

E. References

- Creswell, J. W., & Creswell, J. D. (2018). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. SAGE Publications.

- Dawson, C. (2019). Projects in Computing and Information Systems: A Student's Guide. Pearson Education.
- Oates, B. J. (2006). Researching Information Systems and Computing. SAGE Publications.
- Robson, C., & McCartan, K. (2016). Real World Research. Wiley.
- Sommerville, I. (2016). Software Engineering. Pearson Education.

Lesson 3:

Writing the Review of Related Literature and Studies

A. Learning Objectives

1. Define and explain the importance of the Review of Related Literature and Studies.
2. Differentiate between theoretical/technical background, related literature, and related studies.
3. Identify and organize the elements of a literature review.
4. Apply proper APA in-text citation and referencing techniques.
5. Write a synthesis that justifies the present study.
6. Distinguish between plagiarism and copyright infringement.
7. Demonstrate ethical writing by paraphrasing, summarizing, and citing correctly.

B. Content

The Review of Related Literature and Studies (RRLS) is one of the most important parts of academic and research writing. It provides a foundation for your study by presenting what has already been written, tested, and discovered in your chosen field. By analyzing related works, you ensure that your research does not duplicate past studies unnecessarily but instead builds upon them, filling in gaps or extending knowledge.

In this module, you will learn how to write the different components of RRLS, apply proper citation formats, synthesize ideas, and avoid plagiarism.

The Theoretical or Technical Background

The theoretical background presents established concepts, models, or theories that guide the study. This serves as a framework for understanding and interpreting results.

> **Example (Theoretical):** A study about student motivation may use Maslow's Hierarchy of Needs or Self-Determination Theory.

The technical background, on the other hand, explains the technical aspects such as tools, technologies, or methodologies.

> **Example (Technical):** A project on image processing may explain the basics of neural networks and computer vision.

Together, these provide both conceptual grounding (theory) and operational foundation (technical).

Related Literature

- *Aim of Literature Review*
  - To provide context and background.
  - To identify gaps in existing research.
  - To prevent duplication of work.
  - To justify why the present study is needed.

- *Elements of Literature Review*
  1. Introduction – Defines the scope and coverage.
  2. Body/Discussion – Organized by theme, variable, or time sequence.
  3. Critical Analysis – Evaluates strengths, weaknesses, and limitations.
  4. Synthesis – Connects past studies and links them to the present research.

  Tip: Avoid simply summarizing. Instead, critically analyze and connect ideas.

Related Studies

Before citing sources, it is important to know the two types of in-text citations in APA style:

Citing Literature and Studies (APA Style Examples)

Case	Narrative Citation	Parenthetical Citation
One Author	*Reyes (2020) explained the importance of ICT in education.*	*ICT is important in education (Reyes, 2020).*
Two Authors	*Cruz and Santos (2019) studied blended learning strategies.*	*Blended learning strategies were effective (Cruz & Santos, 2019).*
Three or More Authors	*Gomez et al. (2018) emphasized the role of collaboration.*	*Collaboration is essential (Gomez et al., 2018).*
Group Author – Government Agency	*The Department of Education (DepEd, 2021) implemented new digital literacy programs.*	*Digital literacy programs were implemented (Department of Education [DepEd], 2021).*
Group Author – Organization	*The World Health Organization (WHO, 2020) issued safety guidelines.*	*Safety guidelines were issued (World Health Organization [WHO], 2020).*
No Author	*The report ("Gender Equality Report," 2022) indicated progress in workplace inclusion.*	*Workplace inclusion improved ("Gender Equality Report," 2022).*
Direct Quotation (with page)	*Santos (2021) stated, "ICT reshapes organizational management" (p. 45).*	*"ICT reshapes organizational management" (Santos, 2021, p. 45).*
Direct Quotation (no page, paragraph no.)	*Santos (2021, para. 4) noted the rapid shift to digital platforms.*	*The rapid shift to digital platforms was observed (Santos, 2021, para. 4).*
Secondary Source	*As cited in Dela Cruz (2019), Smith's findings showed consistent patterns.*	*The findings were consistent (Smith, as cited in Dela Cruz, 2019).*

Case	Narrative Citation	Parenthetical Citation
Multiple Works in One Citation	*Reyes (2017), Santos (2018), and Cruz (2020) all contributed to the study of online learning.*	*Several studies explored online learning (Reyes, 2017; Santos, 2018; Cruz, 2020).*

Justification of the Present Study / Project (Synthesis)

The synthesis ties together all reviewed works and shows.

- What has been studied before.
- What remains unexplored.
- How your research addresses the gap.

Example:
"While several studies explored online learning systems (Reyes, 2017; Santos, 2018), few have focused on accessibility for persons with disabilities. This study addresses that gap by designing an inclusive, web-based employment platform tailored for PWDs."

What is Plagiarism?

Plagiarism is using someone else's work, ideas, or words without proper acknowledgment.

Forms of plagiarism:
- Copy-paste without citation.
- Paraphrasing without credit.
- Submitting someone else's work as your own.

Consequences: Academic penalties, loss of credibility, and in some cases, legal action.

Plagiarism vs. Copyright Infringement

Plagiarism	Copyright Infringement
Academic misconduct	Legal violation
Using others' ideas without acknowledgment	Unauthorized reproduction/distribution of copyrighted material
Ethical issue in schools/research	Civil or criminal liability under law
Punished by schools/universities	Punished by law courts

C. Summary

In this module, you learned about the essential components of Chapter III: Writing the Review of Related Literature and Studies. It began with understanding the theoretical and technical background, followed by exploring related literature and related studies. You discovered the aims and elements of a literature review, how to cite sources properly using APA style, and the difference between narrative and parenthetical citations.

The module also guided you in synthesizing information to justify the current study and emphasized the importance of academic honesty by distinguishing plagiarism from copyright infringement. Through activities and assessments, you practiced citation, synthesis writing, and plagiarism prevention strategies.

D. Learning Activities

1. Concept Mapping – Create a diagram showing the link between theoretical background, related literature, and related studies.
2. Source Hunt – Find 3 journal articles related to your research topic and cite them correctly in APA style.
3. Synthesis Writing – Write a paragraph synthesizing three related works about a chosen topic.
4. Plagiarism Check – Paraphrase this statement: “Technology has transformed the way government agencies deliver public services.” (Santos, 2021, p. 30).

Assessment / Quiz

5. What is the main purpose of a literature review?
6. Differentiate between theoretical and technical background.
7. Cite the following correctly: Author – Maria Cruz; Year – 2022; Page – 15.
8. What makes a synthesis different from a summary?
9. Which is an example of plagiarism?
   a. Citing an author with page number
   b. Copy-pasting a paragraph without citation
   c. Writing your own analysis of three articles
   d. Quoting with acknowledgment
10. Give one difference between plagiarism and copyright infringement.
11. Write an in-text citation for three authors: Reyes, Santos, and Cruz (2019).
12. Provide an example of multiple works cited in one statement.
13. Why is it important to analyze, not just summarize, related works?
14. Explain in your own words how RRLS helps justify your study.

• Narrative Citation – The author's name is part of the sentence, followed by the year in parentheses.
Example: Reyes (2020) explained that ICT reshapes organizational management.

• Parenthetical Citation – Both the author's name and year appear in parentheses at the end of the sentence.
Example: ICT reshapes organizational management (Reyes, 2020).

E. References

- American Psychological Association. (2020). Publication manual of the American Psychological Association (7th ed.). APA.
- Boote, D. N., & Beile, P. (2005). Scholars before researchers: On the centrality of the dissertation literature review in research preparation. Educational Researcher, 34(6), 3–15. https://doi.org/10.3102/0013189X034006003
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- Stern, P. N., & Porr, C. J. (2017). Essentials of accessible grounded theory. Routledge.
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Lesson 4:

Writing the Methodology

A. Learning Objectives

At the end of this lesson, students should be able to:

1. **Explain** the importance of the methodology section as the backbone of a thesis or capstone project.
2. **Identify and describe** different types of research designs applicable in IT/IS capstone projects (descriptive, experimental, case study, Agile/Waterfall).
3. **Select and justify** appropriate research instruments for data collection, such as surveys, interviews, or system logs.
4. **Differentiate** between probability and non-probability sampling techniques and **apply them** in IT/IS research contexts.
5. **Summarize and use** CHED's suggested methodology template for IT/IS capstone projects.
6. **Recognize and apply** international standards in software design and development (e.g., ISO/IEC 25010, IEEE standards).
7. **Conduct requirement analysis** by identifying functional and non-functional requirements for a proposed system.
8. **Prepare requirement documentation** (SRS, UML diagrams, ERDs) to define the scope of the system clearly.
9. **Develop an implementation plan** that includes a timeline, testing strategies, deployment, and maintenance.

B. Content

The methodology is the backbone of any thesis or capstone project. It describes the overall approach, processes, and procedures undertaken to complete the study or project. In Information Technology (IT) and Information Systems (IS) research, the methodology often involves software/system development, data collection, testing, and evaluation of the proposed system. This section must be written with clarity so that others can replicate or validate the study.

Research Design, Instrument, and Sampling Techniques

1. **Research Design**

   The research design refers to the structure and strategy of the investigation. It identifies the kind of research undertaken and how the problem will be addressed.

   Examples of Research Design in IT/IS Capstone Projects:
   - **Descriptive Design** – Describes existing systems, user needs, or problems.
   - **Experimental/Developmental Design** – Tests new systems or compares different solutions.
   - **Case Study** – Focuses on a single organization or system in detail.
   - **Agile or Waterfall Models** – Commonly used in software/system development.

2. **Research Instruments**

   Research instruments are the tools used to gather data from respondents. These instruments must be reliable and valid to ensure accuracy.

   Examples of Instruments:
   - **Survey Questionnaires** – Evaluate system usability (e.g., System Usability Scale).
   - **Interview Guides** – Gather expert insights or user feedback.
   - **Observation Checklists** – Record how users interact with the system.
   - **System Logs/Analytics** – Automatically capture user activity data.

3. **Sampling Techniques**

   Sampling techniques refer to the process of selecting respondents or participants for the study. In IT and IS research, choosing the right participants is essential to ensure that the system is tested by appropriate users.

- *Probability Sampling*
    - ✓ **Simple Random Sampling** – Every individual has an equal chance of being chosen.
      Example: Randomly selecting 50 students from a university to test a new e-learning platform.
    - ✓ **Stratified Sampling** – The population is divided into subgroups (strata) and samples are taken proportionally.
      Example: Selecting 20 students from each year level to test an online enrollment system.
    - ✓ **Cluster Sampling** – Dividing the population into clusters and selecting entire clusters randomly.
      Example: Choosing two company branches at random to test a new payroll system.
- *Non-Probability Sampling*
    - ✓ **Purposive Sampling** – Selecting participants with specific characteristics relevant to the study.
      Example: Choosing only doctors and nurses to test a hospital information system.
    - ✓ **Convenience Sampling** – Selecting participants who are easily available.
      Example: Using classmates to test a food delivery application.
    - ✓ **Quota Sampling** – *Ensuring specific groups are represented in the sample.*
      Example: Selecting 50% students, 30% faculty, and 20% staff to evaluate a campus voting system.

Sampling Techniques Summary Table

Type	Method	Description	Example in IT/IS Capstone
Probability Sampling	Simple Random	Every member has equal chance	Selecting 50 random students to test an e-learning platform
	Stratified	Population divided into subgroups	Selecting 20 students from

			each year level for an online enrollment system
	Cluster	Random selection of groups/clusters	Choosing two branches of a company to test a payroll system
Non-Probability Sampling	Purposive	Based on specific characteristics	Choosing doctors and nurses to test a hospital info system
	Convenience	Based on accessibility/ease	Using classmates to test a food delivery app
	Quota	Ensures representation of groups	Selecting 50% students, 30% faculty, 20% staff for campus voting system

CHED's Suggested Template for Methodology (IT/IS Capstone Projects)

According to CHED Memorandum Order (CMO) 25 s.2015, the methodology chapter for IT and IS capstone projects should include:

1. Research Design
2. Development Model (e.g., SDLC, Agile, Waterfall)
3. Project Procedures (Step-by-step activities)
4. Research Instruments
5. Sampling Techniques and Respondents
6. Data Gathering Procedure
7. Data Analysis Plan

Software Development Methodologies

1. Waterfall Model - A linear, sequential model where each phase (requirements, design, implementation, testing, deployment, maintenance) must be completed before moving to the next.

- **Advantages**: Simple, well-documented, easy to manage.
- **Disadvantages**: Rigid, hard to adapt to changes, issues discovered late.
- **Example**: A payroll system where requirements are fixed and unlikely to change.

2. V-Model (Verification and Validation Model) - An extension of Waterfall where every development stage has a corresponding testing stage.

- **Advantages**: Emphasizes testing, ensures early error detection.
- **Disadvantages**: Still rigid, not suitable for changing requirements.
- **Example**: Banking software requiring strict compliance and testing.

3. Iterative Model - Development is done in cycles (iterations), with partial implementation in each cycle until the full system is complete.

- **Advantages**: Early system versions are delivered, feedback is incorporated.
- **Disadvantages**: Requires more resources and planning.
- **Example**: An e-learning system gradually adding modules like quizzes, forums, and analytics.

4. Spiral Model - Combines iterative development with risk analysis in each cycle.

- **Advantages**: Good for high-risk projects, flexible, risk-driven.
- **Disadvantages**: Expensive, complex to manage.
- **Example**: Defense or aerospace software projects.

5. Agile Methodology - A flexible, iterative approach that values collaboration, adaptability, and customer feedback.

- **Advantages**: Fast delivery, user feedback-driven, adaptable.
- **Disadvantages**: Less documentation, may lack predictability.
- **Example**: A mobile app developed with constant client feedback.

6. Scrum (Agile Framework) - A popular Agile framework where work is divided into **sprints** (2–4 weeks), with defined roles (Product Owner, Scrum Master, Development Team).

- **Advantages**: Encourages teamwork, rapid iteration, and transparency.
- **Disadvantages**: Requires experienced teams, frequent client involvement.
- **Example**: Developing a web-based food delivery platform.

7. DevOps Methodology - Integrates development (Dev) and operations (Ops) for continuous integration and deployment (CI/CD).

- **Advantages**: Faster delivery, automation, improved collaboration between developers and IT operations.
- **Disadvantages**: Requires advanced tools and culture change.
- **Example**: Cloud-based applications with regular updates and monitoring.

Standards in Software Evaluation

Software evaluation is a systematic process to determine if a software product meets its requirements and quality expectations. It ensures reliability, usability, and overall performance before deployment.

Standards such as ISO/IEC 25010 and IEEE guidelines provide frameworks for evaluating software quality. Usability evaluation, particularly, is essential in IT/IS capstone projects. The System Usability Scale (SUS) is a widely used tool that measures usability in terms of effectiveness, efficiency, and satisfaction. Incorporating SUS allows developers and researchers to obtain quantitative feedback from end users, helping to improve system design.

The following are the Standards in Software Evaluation:

1. ISO/IEC 25010 – Software Product Quality - Evaluates software across eight characteristics: Functional suitability, Performance efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

- **Example**: Testing a web-based employment platform for PWDs for usability, performance, and security.

2. IEEE Standards - Guides documentation, testing, and evaluation. Common standards include:

- IEEE 829 – Test Documentation
- IEEE 730 – Software Quality Assurance
- IEEE 1012 – Verification & Validation

Example: Preparing a QA report for a capstone system using IEEE 829.

3. Effectiveness, Efficiency, and Satisfaction (ISO 9241-11)

- **Effectiveness** – Accuracy and completeness in task completion.
- **Efficiency** – Resources (time, effort) used to complete tasks.
- **Satisfaction** – User comfort and acceptability of the system.

Example: Evaluating a student portal's registration process for speed, correctness, and ease of use.

4. System Usability Scale (SUS) - A **10-item questionnaire** used to measure users perceived usability of a system. Provides a **quantitative score (0–100)** that reflects overall usability. SUS items are alternately positively and negatively worded; users rate agreement on a 5-point Likert scale.

Example:

- Users of a capstone food delivery app respond to SUS items like:
    - "I think that I would like to use this system frequently."
    - "I found the system unnecessarily complex."
- Scores are calculated to determine usability percentile and benchmark against other systems.

5. Other Evaluation Metrics

- **Functional Metrics** – Feature completeness, correctness, requirement adherence.
- **Non-Functional Metrics** – Response time, scalability, reliability, maintainability.
- **Tools for Measurement** – JMeter, Selenium, SonarQube, or custom surveys.

Example: Using SUS and performance tests to evaluate a campus voting system.

Requirement Analysis

Requirement analysis involves identifying user needs and system requirements. It is the process of identifying, documenting, and validating the needs of users and stakeholders. It involves determining functional and non-functional requirements and translating them into a clear specification that guides system design and development.

- **Functional Requirements -** Define what the system must do and represent specific features, functions, and behavior.

  Examples for Capstone Projects:

  - E-learning System: "The system shall allow students to submit assignments online."
  - Library Management System: "The system shall allow librarians to track borrowed books."

- **Non-Functional Requirements** - Define **how the system performs**, focusing on quality attributes. It includes performance, usability, reliability, security, scalability, and maintainability.

  Examples:

  - E-learning System: "The system shall support 500 concurrent users."
  - Library Management System: "The system shall provide data encryption to secure student records."

Requirement Documentation

All gathered requirements should be documented for clarity.

Common documentation includes:

- **Software Requirement Specification (SRS)** - A formal document that contains all functional and non-functional requirements.
- **Visual Models:**
  - Use Case Diagram – Illustrates system actors and interactions.
  - Activity Diagram – Shows workflow and processes.
  - Entity-Relationship Diagram (ERD) – Represents data entities and their relationships.
  - Class Diagrams (UML) – Shows system classes and attributes for object-oriented design.

The Implementation Plan

The implementation plan outlines how the system will be developed, tested, and deployed. Key components include:

- Development Timeline (Gantt Chart)
- Testing Plan (Unit Testing, User Acceptance Testing)
- Deployment Strategy
- Maintenance Plan

C. Summary

In this module, the methodology explains the step-by-step process of how the study or system is conducted. The research design provides the overall structure, guiding the project in a logical and organized manner. To ensure that the data collected is accurate and relevant, proper research instruments and sampling techniques are used. Sampling can be done through probability methods such as random, stratified, or cluster sampling, or through non-probability methods like purposive, convenience, or quota sampling. CHED also provides a recommended methodology template for IT and IS capstone projects to standardize the process. In addition, software design must follow international standards to ensure quality, reliability, and usability. Requirement analysis and documentation clearly define the scope and specifications of the system, while a well-prepared implementation plan ensures smooth development, testing, and deployment. By following these steps, students can systematically develop a functional, high-quality, and reliable software system.

D. Learning Assessment

Part I: Short Answer

- Explain the difference between functional and non-functional requirements.
- Describe one advantage and one disadvantage of the Agile methodology in capstone projects.
- Name two types of research instruments and explain how they are used in IT/IS research.
- What is the purpose of CHED’s suggested methodology template for IT/IS capstone projects?
- Give one example of a visual model used in requirement documentation and explain its purpose.

Part II: Application / Practical

- Develop a small SRS snippet for a web-based library system including one functional and one non-functional requirement.
- Using SUS, create two sample questions that can measure the usability of a capstone e-learning system.
- Construct a sampling plan for testing a mobile food delivery application with at least one probability and one non-probability technique.
- Draw a simple Use Case Diagram for a student portal system showing at least two actors and two major functions.
- Create a short implementation plan outline for deploying a capstone project, including timeline, testing, and maintenance.

E. References

- Creswell, J. W., & Creswell, J. D. (2018). Research design: Qualitative, quantitative, and mixed methods approach (5th ed.). Sage Publications.
- Kothari, C. R. (2004). Research methodology: Methods and techniques (2nd ed.). New Age International.
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Lesson 5:

Writing the Results and Discussions

A. Learning Objectives

At the end of this module, students should be able to:

1. Present results of system implementation in a structured format.
2. Analyze and interpret results of software development projects.
3. Differentiate between quantitative and qualitative evaluation of system performance.
4. Apply proper techniques in presenting findings (tables, figures, discussions).
5. Write a comprehensive Results and Discussions chapter tailored for IT/IS Capstone Projects.

B. Content

The Results and Discussions chapter is one of the most important parts of a capstone project. It highlights what the system achieved and how the findings are interpreted in the context of the project's objectives. For a software development project, this involves presenting technical results such as system performance, usability testing, accuracy of implemented features, and user evaluation. This section ensures that the technical work is clearly communicated and that readers understand the implications of the results.

Time Signature and Beats Per Minute

The system's ability to detect time signature and beats per minute (BPM) was evaluated using a dataset of sample music files. Results showed that the BPM detection module achieved an accuracy rate of 92%. For instance, a test file with a 4/4-time signature at 120 BPM was correctly identified by the system. This demonstrates the system's reliability in processing rhythm-based features of music.

Melody Extraction Implementation

Melody extraction was tested by analyzing audio tracks and comparing extracted melodies against reference notations. Results revealed that the system accurately extracted the main melody line in 88% of test cases. Errors were observed in tracks with overlapping vocals or background instruments. Nevertheless, the feature proved effective for educational and analytical use.

Chords Detection Implementation

Chord detection was implemented using machine learning models trained on annotated chord datasets. The system correctly identified chords in 85% of the cases. Misclassifications typically occurred in complex jazz progressions, but overall performance remained strong for common chord structures such as major, minor, and seventh chords.

Qualitative Evaluation/Presentation

The qualitative evaluation involved gathering feedback from end-users including music students, faculty, and IT professionals. A Likert-scale questionnaire was used to evaluate ease of use, efficiency, reliability, and overall satisfaction. Results were summarized into mean scores and interpreted qualitatively.

Table 1: *User Evaluation Results (Likert Scale)*

Criteria	Mean Score	Qualitative Interpretation
Ease of Use	4.5	Very Satisfactory
Efficiency (Speed & Accuracy)	4.2	Satisfactory
Reliability (Error Handling)	4.3	Satisfactory
User Interface Design	4.6	Very Satisfactory
Overall Satisfaction	4.4	Very Satisfactory

Scale: 1 – Poor, 2 – Fair, 3 – Good, 4 – Satisfactory, 5 – Very Satisfactory

Table 2: *System Usability Scale (SUS) Results*

Respondent Group	No. of Respondents	SUS Score (Mean)	Interpretation
Students	20	81.2	Excellent
Faculty Evaluators	10	79.5	Good
IT Professionals	5	85.0	Excellent
Overall	35	82.0	Excellent

Interpretation: SUS Score 80+ = Excellent, 70–79 = Good, 60–69 = Acceptable, <60 = Poor

Analysis and Interpretation of the Data

As shown in Table 5.1, the system obtained an overall mean rating of 4.4 (Very Satisfactory) across key usability criteria, indicating that users found the system effective and easy to use. The highest score was observed in User Interface Design (4.6), while the lowest was in Efficiency (4.2), suggesting that while the design was well-received, further improvements could be made in optimizing system speed.

Similarly, the System Usability Scale (Table 5.2) yielded an overall score of 82.0, which falls under the “Excellent” category. IT professionals gave the highest rating (85.0), affirming the system’s technical soundness, while faculty evaluators provided slightly lower scores (79.5), noting areas for refinement in reporting features.

C. Summary

This chapter has presented the results of the developed system, including functional implementation (e.g., time signature detection, melody extraction, chord detection) and user evaluations. Both quantitative results (accuracy, performance scores) and qualitative feedback (usability, user satisfaction) were discussed. Findings indicate that the system is effective and reliable, with minor areas for improvement. Overall, the Results and Discussions chapter validates the project's objectives and demonstrates the system's value to its intended users.

D. Learning Assessment

Answer the following to assess your understanding of this module:

1. Why is it important to present both quantitative and qualitative results in the Results and Discussions chapter?
2. Based on the sample evaluation tables, what areas of the system could still be improved?
3. How can the way results are presented influence the interpretation of findings?
4. Create a sample results table for an imaginary feature (e.g., genre classification, log-in success rate) and provide a short interpretation.
5. Explain how the Results and Discussions chapter serves as validation of the project objectives.

E. References

- Creswell, J. W., & Creswell, J. D. (2018). *Research design: Qualitative, quantitative, and mixed methods approach* (5th ed.). Sage Publications.
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Lesson 6:

How to Write the Summary, Conclusions, and Recommendations of the Study

A. Learning Objectives

At the end of this module, students should be able to:

1. Differentiate between a summary, conclusions, and recommendations.
2. Write a concise and accurate summary of findings and accomplishments.
3. Formulate logical conclusions based on presented results.
4. Develop actionable recommendations aligned with the study's objectives.
5. Distinguish between an abstract and an executive summary.

B. Content

The final chapter of a thesis or capstone project is critical because it synthesizes the entire research work. It serves as the culmination of the study, presenting a **summary of findings**, drawing **conclusions**, and suggesting **recommendations** for practice, policy, or further research. Additionally, researchers are often required to write an **abstract** or **executive summary** for quick reference.

This chapter teaches students how to clearly and effectively summarize their research, provide evidence-based conclusions, and propose practical recommendations.

Qualities of a Summary of Findings and Accomplishments

A good summary should:

- Be **concise and direct**.
- Present only the **key results** (not raw data).
- Be written in **past tense**.
- Highlight accomplishments in relation to the objectives.

Example:
"The system successfully implemented an automatic chord detection feature with an accuracy rate of 92%. Additionally, usability testing with 30 participants indicated that the system achieved a satisfaction rating of 4.5 out of 5."

Conclusions

Conclusions are the logical interpretation of findings. They should answer the research questions and validate (or invalidate) the objectives of the study.

Qualities of Good Conclusions:

- Directly derived from findings.
- Free of new data.
- Answer the objectives of the study.
- Stated in a clear, conclusive manner.

Example:
"The results confirm that the system is effective in detecting melody and chords with acceptable accuracy, making it suitable for music education and personal practice."

Summary vs. Conclusions

Aspect	Summary of Findings	Conclusions
Content	What was discovered (facts and results).	What the results mean (interpretation).
Focus	Objective-based accomplishments.	Implications of the findings.
Style	Descriptive.	Interpretative and evaluative.

Recommendations

Recommendations are actionable suggestions based on the conclusions of the study. They may be directed toward:

- **Future researchers** – suggestions for further studies.
- **Practitioners/Users** – improvements in the actual use of the system.
- **Developers** – technical enhancements.

  Example:
  "It is recommended to expand the system's training dataset to improve the accuracy of chord detection. Future researchers may also integrate genre classification to enhance system functionality."

The Abstract and Executive Summary

Both provide a concise overview but differ in scope and purpose:

- **Abstract**: Short (150–250 words), academic, focused on research problem, methods, results, and conclusion.
- **Executive Summary**: Longer (1–2 pages), practical, designed for decision-makers, includes context, findings, conclusions, and recommendations.

The Abstract vs. Executive Summary

Aspect	Abstract	Executive Summary
Length	150–250 words.	1–2 pages.
Audience	Academic community, researchers.	Policymakers, managers, practitioners.

Aspect	Abstract	Executive Summary
Content	Problem, objectives, methods, findings, conclusion.	Context, findings, conclusions, recommendations.
Tone	Academic and technical.	Professional and practical.

C. Summary

This chapter highlighted the importance of effectively summarizing findings, drawing logical conclusions, and providing relevant recommendations. It also clarified the differences between a summary and conclusions, and between an abstract and an executive summary. These skills ensure that a research project communicates its contributions clearly and persuasively.

D. Learning Assessment

1. Differentiate between a **summary of findings** and **conclusions** in your own words.
2. Provide an example of a **recommendation** for a capstone project on **music recognition software**.
3. Why is it important that conclusions should not introduce new data?
4. Compare an **abstract** and an **executive summary** in terms of length, purpose, and audience.
5. Draft a sample **abstract** (150–200 words) based on a hypothetical project about an AI-powered web platform for job matching.

E. References

- Creswell, J. W., & Creswell, J. D. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). Sage Publications.
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Lesson 7:

Writing Using ACM Template

A. Learning Objectives

After completing this module, students should be able to:

- Understand the importance of using the ACM (Association for Computing Machinery) template in research writing.
- Identify the guidelines and requirements when preparing a paper in ACM format.
- Recognize the format and structure of an ACM paper, including major sections.
- Apply ACM formatting standards in writing their capstone project manuscripts.

B. Content

The Association for Computing Machinery (ACM) is one of the world's largest computing societies and a leading publisher of research in computer science, information technology, and related fields.

To ensure uniformity, professionalism, and readability, ACM has developed a standard writing template used for conference and journal publications. For students in computing disciplines, particularly those writing a capstone project or thesis, learning how to use the ACM template is essential because it prepares them for professional research publication.

This chapter introduces the guidelines, format, and structure of the ACM template and provides examples on how to adapt it for academic projects.

Guidelines in Writing ACM Template

1. **Document Preparation**
    - Use the official ACM template in Word or LaTeX.
    - Maintain the specified font, margins, and spacing.
2. **Content Guidelines**
    - Write in clear, concise, and formal language.
    - Include a descriptive title, relevant keywords, and follow ACM's citation style.
3. **Submission Readiness**
    - Ensure grammatical correctness.
    - Avoid plagiarism and ensure originality.
    - Properly label figures and tables.
    - Format bibliography in ACM citation style.

Format and structure of ACM Template

A standard ACM paper follows this structure:

- **Title and Author Information** – Concise title, author(s), affiliation(s), and email(s).
- **Abstract** – 150–250 words summarizing the problem, method, results, and conclusion.

- **CCS Concepts and Keywords** – Select relevant Computing Classification System (CCS) categories and include 3–5 keywords.
- **Introduction** – Presents background, motivation, problem, and objectives.
- **Related Work / Literature Review** – Summarizes previous studies and research gaps.
- **Methodology / System Design** – Describes design, tools, and procedures (e.g., architecture, diagrams, algorithms).
- **Results and Discussion** – Presents findings with text, tables, and figures; provides analysis.
- **Conclusion and Future Work** – Summarizes contributions and suggests improvements.
- **Acknowledgments** – Recognize funding, institutions, or individuals.
- **References** – Follow ACM numeric citation format.

Example (Excerpt in ACM Style)

- **Narrative Citation:**
  As demonstrated by Cruz et al. [2], context-aware filtering improves recommendation accuracy.
- **Parenthetical Citation:**
  Context-aware filtering has been shown to improve recommendation accuracy [2].

C. Summary

This chapter provided an overview of how to write a research paper using the ACM template. The guidelines emphasize proper formatting, clarity of writing, and adherence to ACM's structured format. Students preparing capstone projects are encouraged to use this template not only to comply with academic requirements but also to prepare their work for possible publication in ACM-affiliated conferences and journals.

D. Learning Assessment

Answer the following questions to check your understanding:

1. What is the primary purpose of using the ACM template in research writing?
2. List the three main parts of an ACM paper that appear before the Introduction.
3. Differentiate between ACM's citation style and APA's citation style.

4. Provide an example of an ACM-formatted reference entry for a journal article.
5. Why is it important for computing students to learn ACM formatting at the capstone level?

E. References

- Association for Computing Machinery (ACM). (2023). *ACM Master Article Template.* Retrieved from https://www.acm.org/publications/proceedings-template
- ACM. (2022). *ACM Reference Format Guidelines.* Retrieved from https://www.acm.org/publications/authors/reference-format
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